



ESP-IDF

This chapter contains the following sections. Please read as needed:

- [ESP-IDF Getting Started](#)
- [Setting Up Development Environment](#)
- [Example](#)

ESP-IDF Getting Started

New to ESP32 ESP-IDF development and looking to get started quickly? We have prepared a general [Getting Started Tutorial](#) for you.

- [Section 1: Environment Setup](#)
- [Section 2: Running Examples](#)
- [Section 3: Creating a Project](#)
- [Section 4: Using Components](#)
- [Section 5: Debugging](#)
- [Section 6: FreeRTOS](#)
- [Section 7: Peripherals](#)
- [Section 8: Wi-Fi Programming](#)
- [Section 9: BLE Programming](#)

Please Note: This tutorial uses the [ESP32-S3-Zero](#) as a teaching example, and all hardware code is based on its pinout. Before you start, it is recommended that you check the pinout of your development board to ensure the pin configuration is correct.

Setting Up Development Environment

! INFO

For the ESP32-S3-CAM-OVxxxx development board, ESP-IDF version V5.5.1 or above is required.

NOTE

The following guide uses **Windows** as an example, demonstrating development using **VS Code + the ESP-IDF extension**. macOS and Linux users should refer to the [official documentation](#).

VERSION SELECTION

The screenshots in this section use **ESP-IDF V5.5.2** as an example. When installing, please select the ESP-IDF version that matches your board's example.

Install the ESP-IDF Development Environment

1. Download the installation manager from the [ESP-IDF Installation Manager](#) page. This is Espressif's latest cross-platform installer. The following steps demonstrate how to use its offline installation feature.

Click the **Offline Installer** tab on the page, then select **Windows** as the operating system and **the ESP-IDF version you need** (the version shown in the screenshot is for reference only — choose the version that fits your actual needs).

ESP-IDF Installation Manager

Installation manager for ESP-IDF

Version v0.7.1

Online Installer

1

Offline Installer

Offline Builder Tools

⚠ Warning: Offline installer files are large (3-4GB). Due to browser memory limitations, you will receive separate installer and archive files to use together.

Offline Installer Creator

Installer Type

2

GUI

CLI

Platform

3

windows-x64

IDF Version

4

v5.5.2

5

Download

After confirming your selection, click the download button. The browser will automatically download two files: the **ESP-IDF Offline Package (.zst)** and the **ESP-IDF Installer (.exe)**.



archive_vv5.5.2_
windows-x64.zst

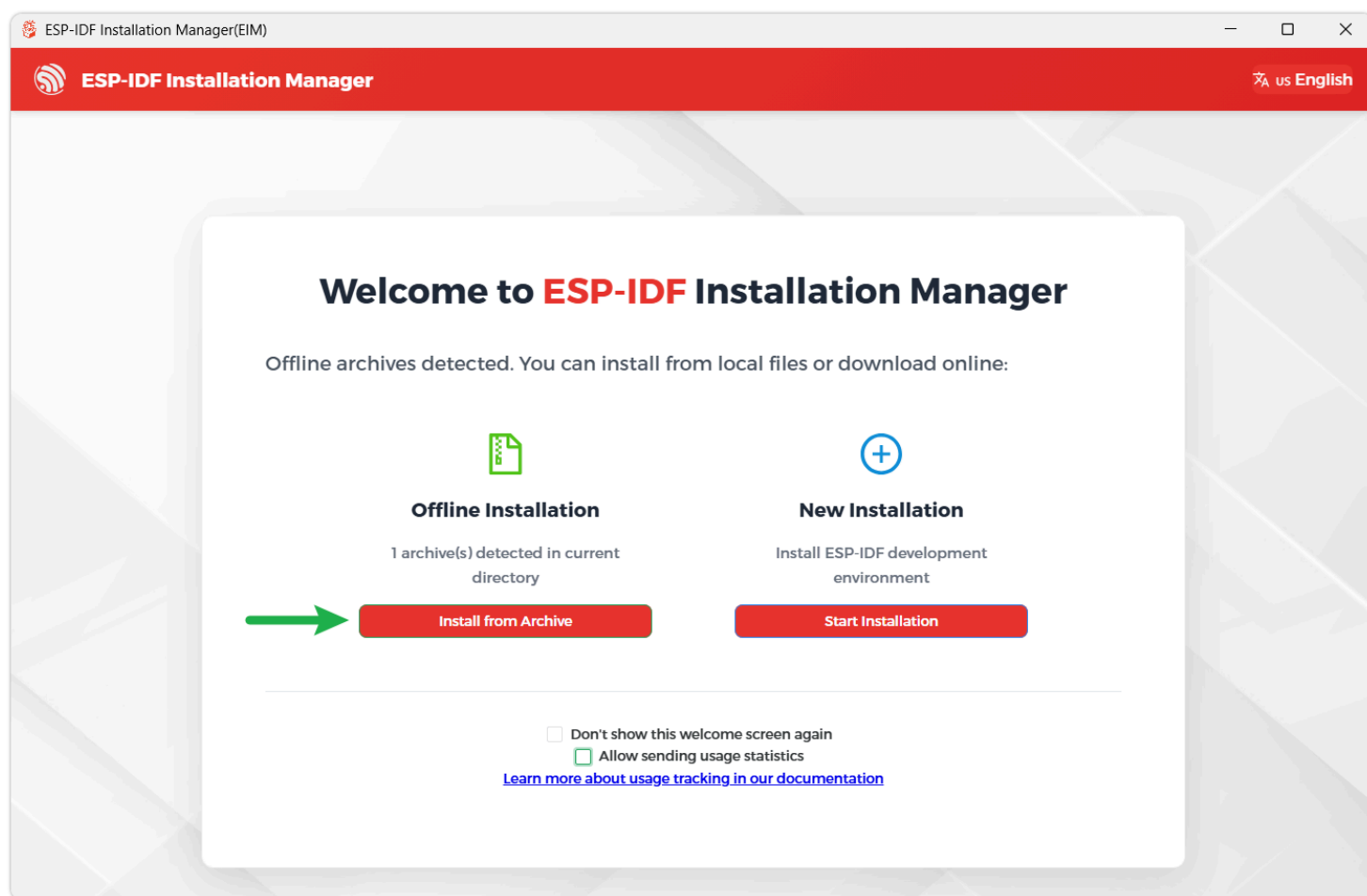


eim-gui-window
s-x64.exe

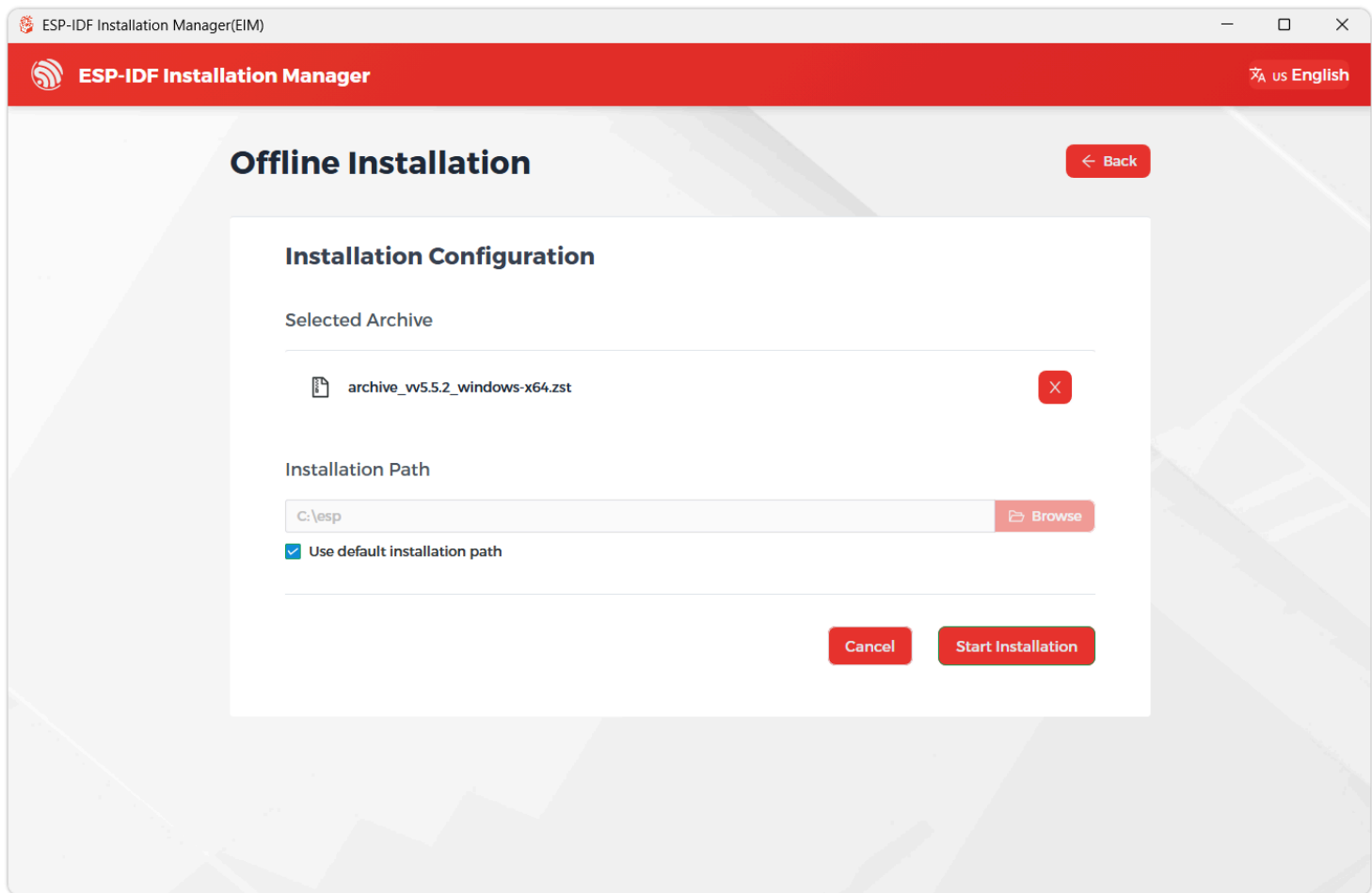
Please wait for both files to finish downloading.

2. Once the download is complete, double-click to run the **ESP-IDF Installer (eim-gui-windows-x64.exe)**.

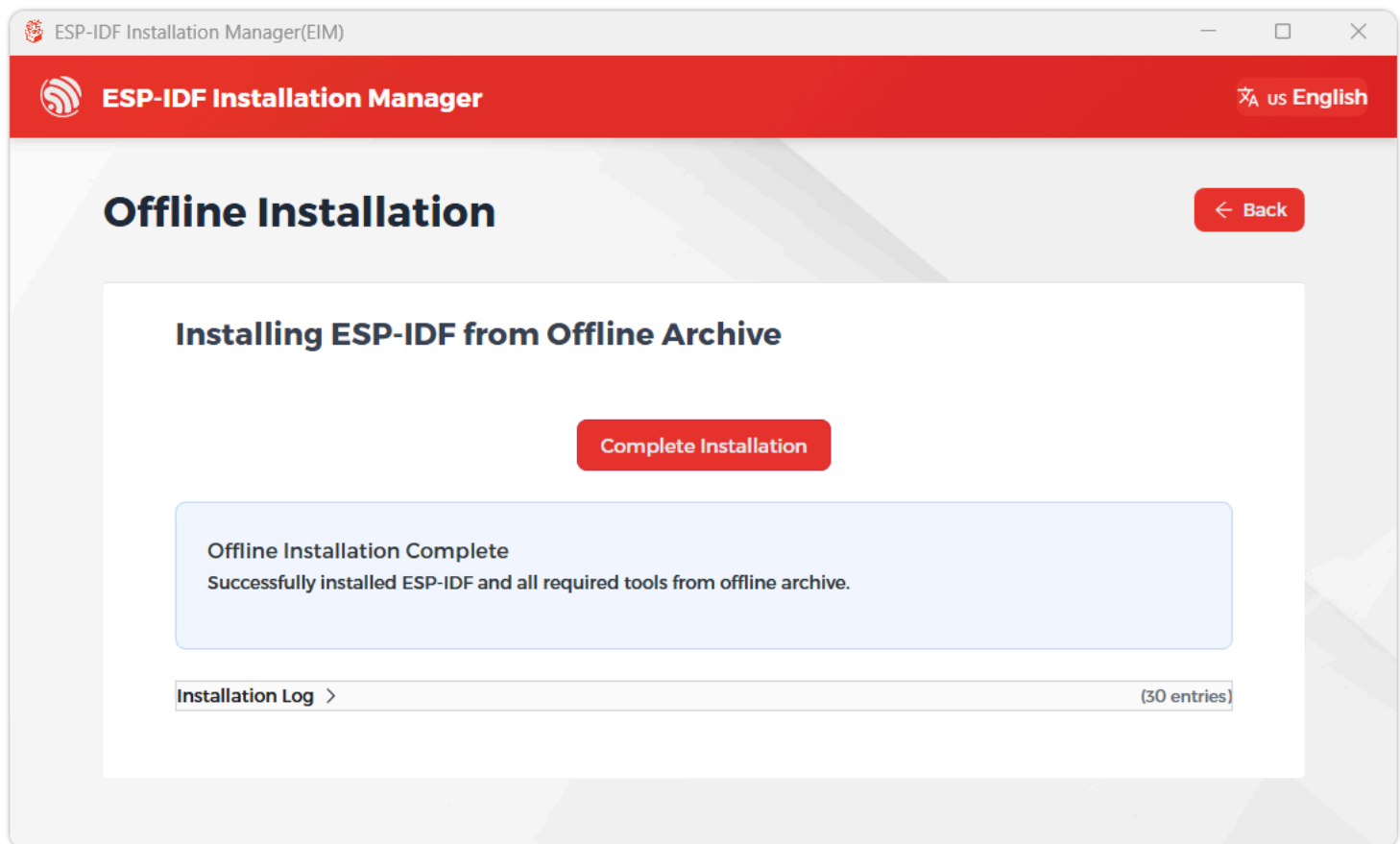
The installer will automatically detect if the offline package exists in the same directory. Click **Install from archive**.



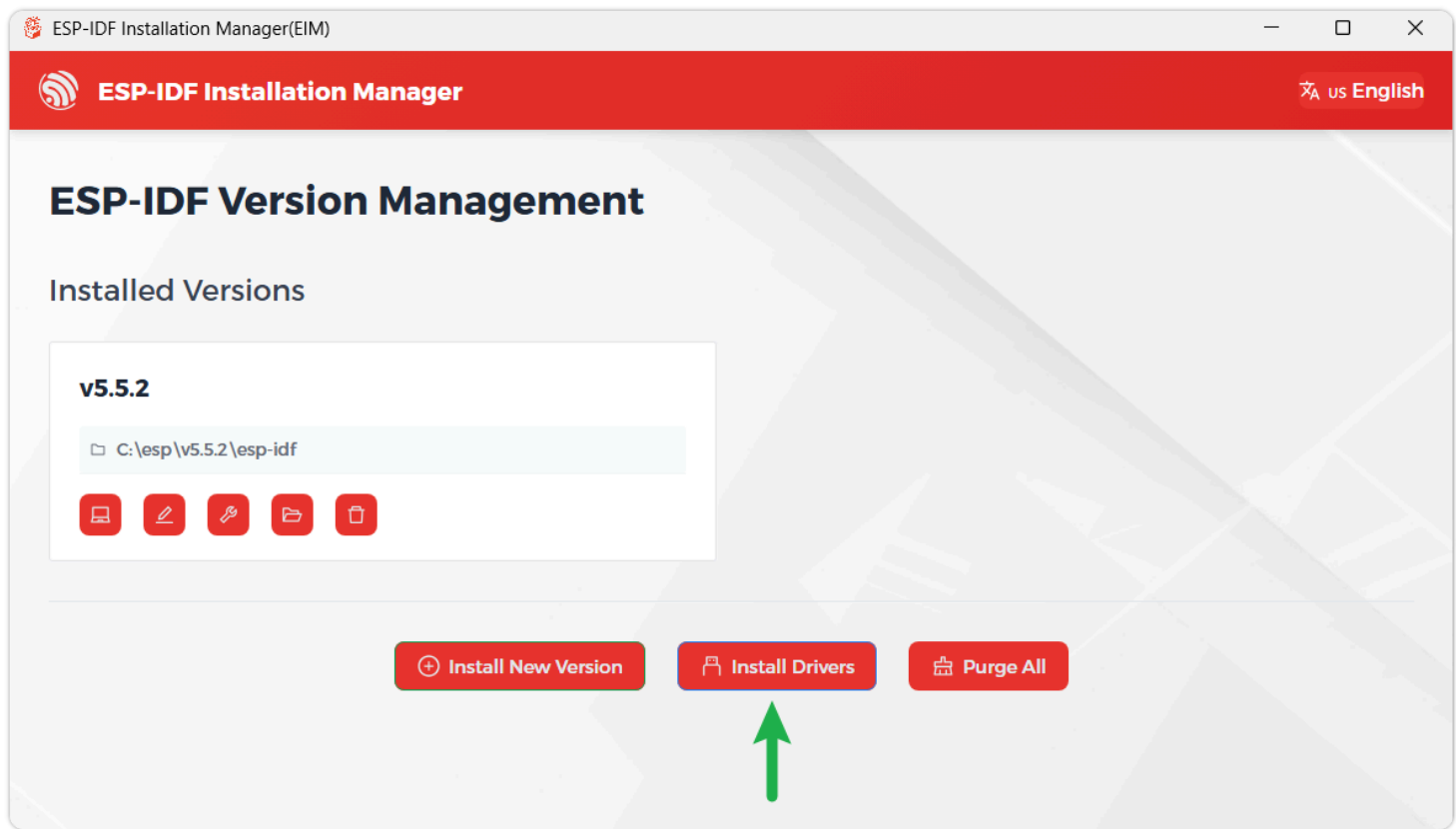
Next, select the installation path. We recommend using the default path. If you need to customize it, ensure the path does not contain Chinese characters or spaces. Click **Start installation** to proceed.



3. When you see the following screen, the ESP-IDF installation is successful.

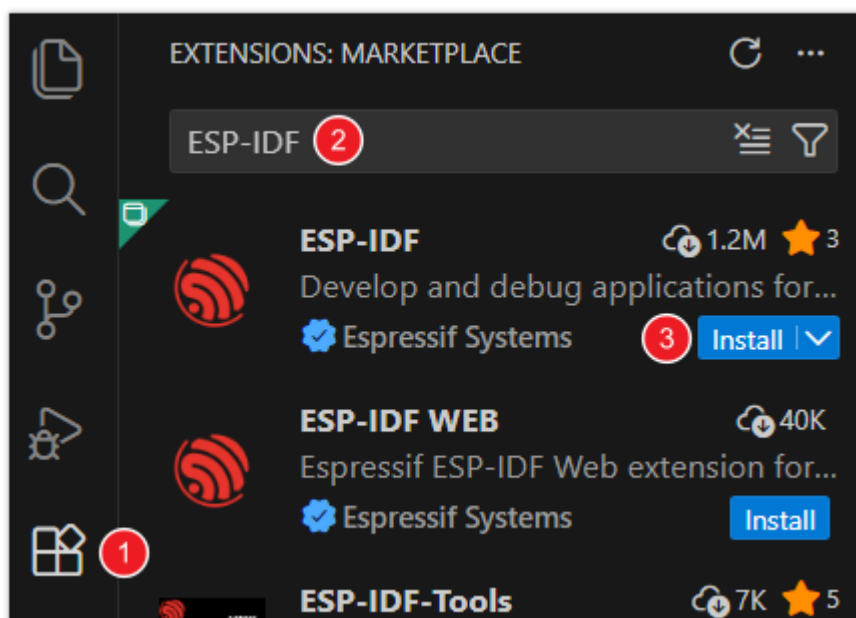


4. We recommend installing the drivers as well. Click **Finish installation**, then select **Install driver**.



Install Visual Studio Code and the ESP-IDF Extension

1. Download and install [Visual Studio Code](#).
2. During installation, it is recommended to check **Add "Open with Code" action to Windows Explorer file context menu** to facilitate opening project folders quickly.
3. In VS Code, click the **Extensions** icon  in the Activity Bar on the side (or use the shortcut `Ctrl + Shift + X`) to open the **Extensions** view.
4. Enter **ESP-IDF** in the search box, locate the [ESP-IDF](#) extension, and click Install.



5. For **ESP-IDF extension versions ≥ 2.0** , the extension will automatically detect and recognize the ESP-IDF environment installed in the previous steps, requiring no manual configuration.

Example

The ESP-IDF examples are located in the `ESP-IDF` directory of the [example package](#).

Demo	Basic Description
01_simple_video_server	Create multiple HTTP servers on different ports on the local network, accessible via a web browser, providing video streaming functionality
02_esp_sr	Demonstrate speech recognition and voice wake-up functions using the ESP-SR component
03_audio_play	Scan the TF card and play MP3 audio
04_dvp_camera_display	Display the camera feed on the LCD screen
05_lvgl_brookesia	Show LVGL APP style UI using the Brookesia component
06_usb_host_uvc	Simulate a UVC device, demonstrating USB camera functionality

01_simple_video_server

Example Description

- This example demonstrates creating an HTTP server, accessing it via a browser, and obtaining a video stream

Hardware Connection

- Connect the development board to the computer
- Camera: Supports GC0308, GC2145, OV5640, OV3660

Code Analysis

- `example_connect()`: Initialize Wi-Fi connection
 - Set SSID and password in the menuconfig configuration
 - Default SSID: WSTEST

- Default password: waveshare0755
- `start_cam_web_server()`: Initialize the HTTP server and camera
 - Initialize the camera first. If an unsupported camera or format is detected, an error will be reported and the program will exit
 - Wait for the camera to initialize successfully, then initialize the HTTP server

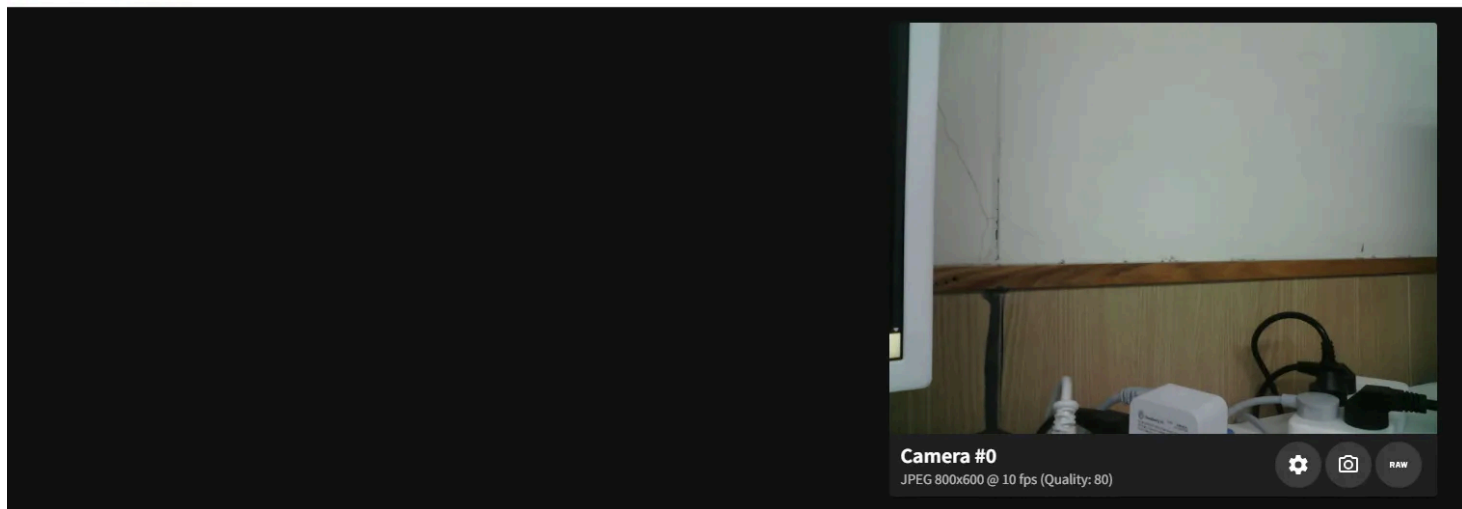
Operation Result

- This demo will not light up the screen
- Check the serial monitor for the printed IP address
- Open the IP address in a browser. Ensure you are on the same local network.

```
I (3145) wifi:dp: 1, bi: 102400, li: 3, scale listen interval from 307200 us to 307200 us
I (3145) wifi:set rx beacon pti, rx_bcn_pti: 0, bcn_timeout: 25000, mt_pti: 0, mt_time: 10000
I (3203) wifi:AP's beacon interval = 102400 us, DTIM period = 3
I (4183) esp_netif_handlers: example_netif_sta ip: 192.168.137.102, mask: 255.255.255.0, gw: 192.168.137.1
I (4183) example_connect: Got IPv4 event: Interface "example_netif_sta" address: 192.168.137.102
I (4580) example_connect: Got IPv6 event: Interface "example_netif_sta" address: fe80:0000:0000:0000:226e:f1ff:fe9a:0fe4, type: ESP_IP6_ADDR_IS_LINK_LOCAL
I (4581) example_common: Connected to example_netif_sta
I (4581) example_common: - IPv4 address: 192.168.137.102,
I (4581) example_common: - IPv6 address: fe80:0000:0000:0000:226e:f1ff:fe9a:0fe4, type: ESP_IP6_ADDR_IS_LINK_LOCAL
I (4787) example: video0: width=320 height=240 format=422P
I (4789) example: Starting stream server on port: '80'
I (4791) example: Camera web server starts
I (4792) main_task: Returned from app_main()
I (5089) wifi:<ba-add>idx:0 (ifx:0, d2:57:7e:c7:43:ea), tid:0, ssn:12, winSize:64
```

192.168.137.74

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02_esp_sr

Example Description

- This example demonstrates offline voice wake-up and speech recognition using the ESP-SR component

Hardware Connection

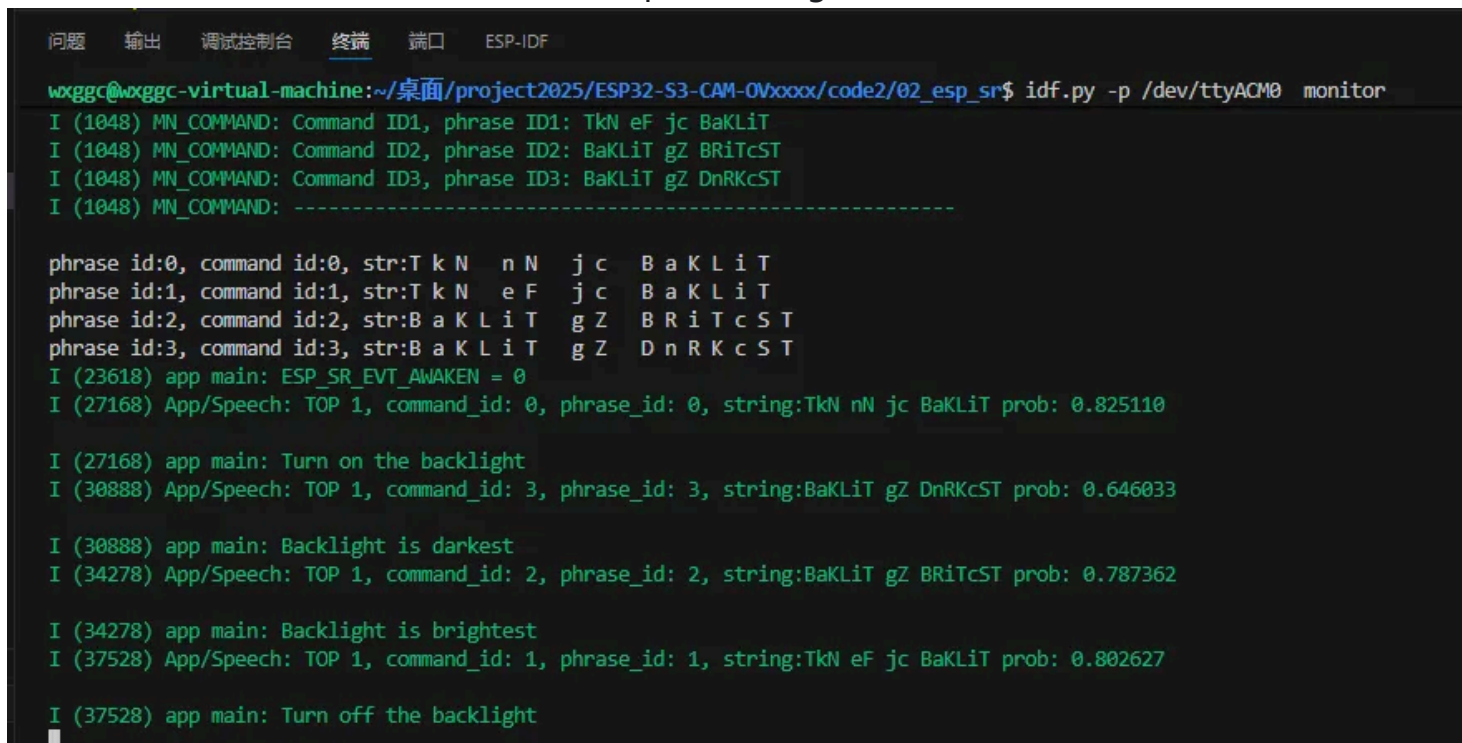
- Connect the development board to the computer
- Do not block the microphone

Code Analysis

- `Speech_Init()`: Initialize the microphone and create recognition tasks
 - Obtain the microphone control handle via `bsp_audio_codec_microphone_init()`
 - Create data reading and detection/recognition tasks
- `Speech_register_callback()`: Register a callback function for recognition events
 - Events can be obtained within the callback function

Operation Result

- This demo will not light up the screen
- Check the serial monitor for voice wake-up and recognition results



```
wxggc@wxggc-virtual-machine:~/桌面/project2025/ESP32-S3-CAM-0Vxxxx/code2/02_esp_sr$ idf.py -p /dev/ttyACM0 monitor
I (1048) MN_COMMAND: Command ID1, phrase ID1: TkN eF jc BaKLiT
I (1048) MN_COMMAND: Command ID2, phrase ID2: BaKLiT gZ BRiTcST
I (1048) MN_COMMAND: Command ID3, phrase ID3: BaKLiT gZ DnRKcST
I (1048) MN_COMMAND: -----

phrase id:0, command id:0, str:T k N   n N   j c   B a K L i T
phrase id:1, command id:1, str:T k N   e F   j c   B a K L i T
phrase id:2, command id:2, str:B a K L i T   g Z   B R i T c S T
phrase id:3, command id:3, str:B a K L i T   g Z   D n R K c S T
I (23618) app main: ESP_SR_EVT_AWAKEN = 0
I (27168) App/Speech: TOP 1, command_id: 0, phrase_id: 0, string:TkN nN jc BaKLiT prob: 0.825110

I (27168) app main: Turn on the backlight
I (30888) App/Speech: TOP 1, command_id: 3, phrase_id: 3, string:BaKLiT gZ DnRKcST prob: 0.646033

I (30888) app main: Backlight is darkest
I (34278) App/Speech: TOP 1, command_id: 2, phrase_id: 2, string:BaKLiT gZ BRiTcST prob: 0.787362

I (34278) app main: Backlight is brightest
I (37528) App/Speech: TOP 1, command_id: 1, phrase_id: 1, string:TkN eF jc BaKLiT prob: 0.802627

I (37528) app main: Turn off the backlight
```

03_audio_play

Example Description

- This example demonstrates playing MP3 audio files from a TF card

Hardware Connection

- Insert a memory card with an MP3 file placed in the root directory
- Connect a speaker
- Connect the development board to the computer

Code Analysis

- `bsp_io_expander_init()`: Initialize the external I/O expander

- Initialize the PA control pin to control audio output
- `bsp_sdcard_mount()`: Mount the TF card
- `Search_Music()`: Search for and save the paths of MP3 files on the TF card
- `Audio_Play_Init()`: Initialize the audio player
- `Volume_Adjustment()`: Set the volume level
- `Audio_Play_Music()`: Play the MP3 file

Operation Result

- This demo will not light up the screen
- After running, it will automatically search for MP3 files on the TF card and play them

```
I (781) ESP_GMF_TASK: Waiting to run... [tsk:TSK_0x3fceb80-0x3fceb80, wk:0x3c12d22c, run:0]
I (781) ESP_GMF_FILE: Open, dir:1, uri:file:///sdcard/music/FF-16B~1.MP3
I (781) ESP_GMF_FILE: File size: 1497402 byte, file position: 0
I (781) ESP_GMF_TASK: One times job is complete, del[wk:0x3c12d22c, ctx:0x3c12cd34, label:aud_dec_open]
I (791) ESP_GMF_PORT: ACQ IN, new self payload:0x3c12d22c, port:0x3c12d184, el:0x3c12cd34-aud_dec
I (791) main_task: Returned from app_main()
```

04_dvp_camera_display

Example Description

- This example is an app-style UI + camera demo. Tapping the camera app displays the image on the screen

Hardware Connection

- Connect a DVP camera: Supports GC0308, GC2145, OV5640, OV3660
- Connect a screen, supports 1.83, 2inch, 2.8inch, 3.5inch
- Connect the development board to the computer

Code Analysis

- `bsp_display_start()`: BSP initializes display-related resources
 - Initialize the LCD screen
 - Initialize LVGL
- `bsp_display_backlight_on()`: Turn on the backlight
- `new ESP_Brookesia_Phone(dispatcher)`: Initialize the Brookesia component
- `new PhoneCameraConf(1,0);`: Install the camera app

Operation Result

- Tap the camera app icon, the screen starts displaying the camera image. Tap anywhere on the screen to return to the camera settings interface





05_lvgl_brookesia

Example Description

- This example demonstrates a complete phone-style UI system, including components such as a status bar, navigation bar, app launcher, and gesture interaction

Hardware Connection

- Connect a screen, supports 1.83, 2inch, 2.8inch, 3.5inch
- Connect the development board to the computer

Code Analysis

- `bsp_display_start()`: BSP initializes display-related resources
 - Initialize the LCD screen
 - Initialize LVGL

Operation Result

- The display effect is shown on a 2inch screen



06_usb_host_uvc

Example Description

- This example simulates a UVC device, demonstrating USB camera functionality

Hardware Connection

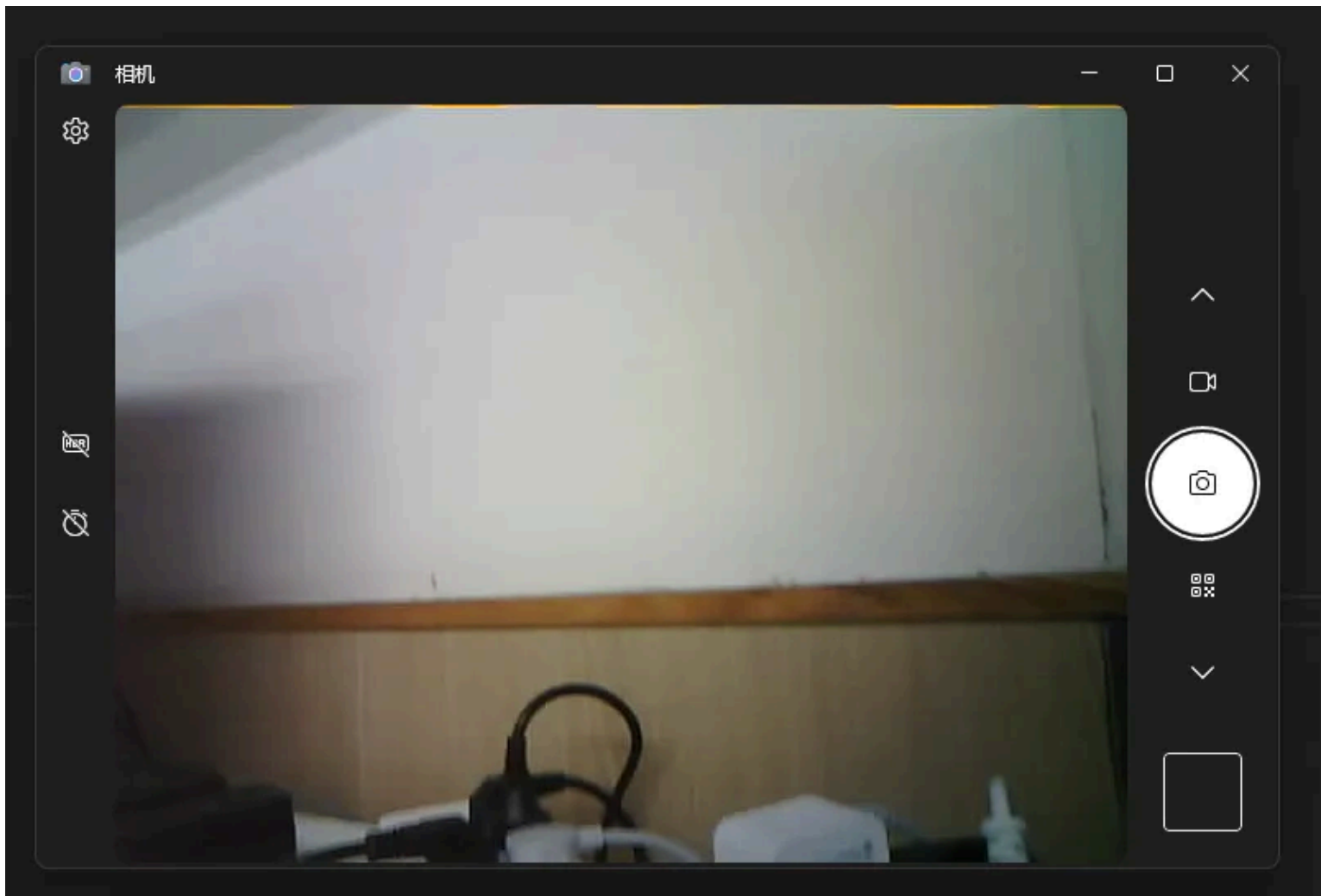
- Connect a DVP camera: Supports OV5640, OV3660
- Connect the development board to the computer

Code Analysis

- `uvc_device_config()`: Configure UVC device-related interface functions
- `uvc_device_init()`: Start initializing the UVC device

Operation Result

- Open the camera app on your computer to see the image. There may be some latency; wait a moment or try re-powering
- This demo uses the USB programming interface. When programming again, you need to hold the BOOT button to enter download mode



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